

Saarland University  
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Master's Thesis

# Optimization-Based Precipitation Estimation from Commercial Microwave Links

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# Abstract

Commercial microwave links provide an opportunistic source of rainfall information: rain attenuates the transmitted signal along each link path, and these attenuation measurements can be used to infer rainfall. Many existing CML rainfall-mapping approaches rely on link-level rain-rate estimates followed by spatial interpolation. In this thesis, we study optimization-based rainfall reconstruction from commercial microwave links as an alternative to purely interpolation-based map construction.

We construct a 100-patch benchmark from radar-derived European rainfall fields, simulate link attenuations using a physical attenuation model, and compare inverse-distance weighting (IDW), a link-aware interpolation variant (ILDW), and several optimization-based reconstructions. These include an ILDW-initialized nonlinear solver, a convex reformulation, a homotopy variant, and a ground-truth-initialized diagnostic solver. The evaluation compares pixelwise agreement with the radar-derived rainfall fields, distance-conditioned errors relative to link proximity, and attenuation consistency under the same forward model used by the solvers.

The results show that optimization-based reconstructions can substantially improve over IDW in patch-level RMSE and Pearson correlation, while the convex and homotopy solvers also match the simulated link attenuations more closely. At the same time, the case studies and attenuation-mismatch results show that minimizing the link-based objective is not always equivalent to recovering the radar-derived ground truth pixel by pixel. This highlights both the promise and the difficulty of formulating rainfall reconstruction from microwave-link measurements as an inverse optimization problem.

# Acknowledgments



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## Chapter 1

# Introduction

Rainfall is one of the most important variables in meteorology, yet measuring it accurately over a geographic area remains difficult. One approach that has been studied extensively is to use commercial microwave links as an indirect source of rainfall observations (Messer et al., 2006; Goldshtein et al., 2009; Chwala and Kunstmann, 2019; Lian et al., 2022). These links, originally installed for telecommunications, are attractive because they are already widespread, operate continuously, and provide measurements that carry information about rainfall without requiring a dedicated rainfall-sensor network. Because rain attenuates the transmitted signal along the link path, attenuation measurements collected on such links can be used to infer rainfall.

Reconstructing rainfall fields from microwave-link data has already been studied in the literature, often through interpolation-based methods that first convert each link into a representative rain-rate value and then spread those values across a region. Simple interpolation rules such as inverse-distance weighting (IDW) are attractive because they are straightforward to implement and provide a practical baseline for rainfall mapping (Lian et al., 2022).

In this thesis, we study whether rainfall reconstruction can be improved by replacing purely interpolative map construction with optimization-based inverse reconstruction. Rather than treating the rainfall map only as an interpolation problem, we formulate reconstruction as an optimization problem that balances agreement with the observed link attenuations against terms that encourage spatially coherent rainfall fields. We evaluate several solver variants derived from this formulation and compare them against IDW, a standard interpolation baseline.

To test this question, we construct a 100-patch benchmark from radar-derived rainfall fields. We then simulate microwave-link attenuations with a physical model relating rainfall to link attenuation and compare IDW, ILDW, and several solver variants derived from the optimization-based formulation.

For reproducibility, the code, experiment configuration files, and generated report artifacts used for this thesis are available in the project repository.

<https://github.com/guyeven/Rainfall-Estimation>.

With this setup in place, the remainder of the thesis reviews the background and prior work, presents the benchmark and methods, reports the empirical comparison, and closes with a discussion of the main conclusions and limitations.

## Chapter 2

# Background and Related Work

In most settings, rainfall is measured primarily with rain gauges and weather radar. Rain gauges provide accurate measurements at their installation sites, but gauge networks are often sparse across a region (McKee and Binns, 2016). Radar gives broad spatial coverage, but it does not measure rain rate directly and must infer it from electromagnetic echoes. This is one reason rainfall estimation remains challenging even where operational radar and gauge networks are already in place (Olsson et al., 2025).

**Weather radar: strengths and limitations.** A weather radar is a sensing system that monitors precipitation remotely by transmitting electromagnetic pulses and recording the energy that is scattered back toward the antenna. The return time gives distance, the antenna pointing gives direction, and the returned signal strength gives reflectivity. Rainfall is then inferred from that reflectivity through a conversion model. This is powerful because a single radar can observe precipitation over a wide region with high spatial and temporal resolution. At the same time, the inference is imperfect. The returned energy is influenced not only by rain, but also by the drop-size distribution and, in some situations, by snow, hail, insects, birds, terrain clutter, and other non-rain scatterers. Operational radar processing therefore includes filtering, classification, and calibration steps before reflectivity is turned into rainfall products.

**Commercial microwave links as opportunistic sensors.** A commercial microwave link is a fixed telecommunication path between two antennas, typically mounted on cell-phone towers. Rain along that path attenuates the transmitted microwave signal. By comparing the received signal during rainy periods with its dry-weather baseline, one can estimate the attenuation caused by rain along the link and then convert that attenuation into a path-averaged rain-rate estimate. The physical attenuation–rainfall relationship itself goes back much earlier, including classical work such as Olsen et al. (1978). The specific idea of using measurements from wireless communication networks for rainfall monitoring is usually traced to Messer et al. (2006). Since then, the topic has been studied extensively and is now a well-established opportunistic-sensing approach for rainfall observation (Goldshtein et al., 2009; Chwala and Kunstmann, 2019; Olsson et al., 2025).

**What information a link carries.** In the present context, a link is modeled geometrically as the line segment connecting its two endpoints. Besides those endpoint coordinates, each link carries a carrier frequency and a polarization. The frequency determines the microwave band at which the signal is transmitted. The polarization describes the orientation of the electric field of the transmitted wave, typically horizontal or vertical. Both frequency and polarization matter because the ITU attenuation coefficients depend on them: the same rain rate leads to different specific attenuation at different frequencies and polarizations.

**From link attenuation to rainfall maps.** Recovering a spatial rainfall field from microwave links is harder than recovering a single path-averaged rain rate. Each link only measures information integrated along a line segment, so many different rainfall fields can be compatible with similar link observations. In practice, this has often led to simple but scalable map-construction strategies. A common approach is to turn each link into a representative rainfall value and then interpolate

those values across space, for example with Kriging or inverse-distance weighting (IDW) (Chwala and Kunstmann, 2019; Olsson et al., 2025). These methods are attractive because they are easy to implement, computationally light, and well suited to larger networks. In the present thesis, this motivates the use of IDW as the main interpolation baseline. We also evaluate ILDW, a link-aware modification of IDW used in this work. Unlike the standard midpoint representation of a CML as a virtual rain gauge, ILDW weights each grid cell by its distance to the full link segment and assigns the link-derived path-average rain rate along that segment.

**What has been done before.** Early signal-processing work already showed that measurements from commercial telecommunication links can be converted into useful rain-rate estimates over two-dimensional domains (Goldshtein et al., 2009). Later studies have tested these ideas on larger link networks and in settings closer to operational rainfall monitoring. Roversi et al., for example, applied the RAINLINK workflow to 357 links in northern Italy and compared the resulting rainfall products against the operational products of the regional weather service, finding that CML-based estimates can complement radar products (Roversi et al., 2020). More broadly, Olsson et al. review opportunistic rainfall observations from the perspective of societal benefit and conclude that the scientific support for their use is already strong, while challenges in access, quality assurance, and operational integration still remain (Olsson et al., 2025). The present thesis fits into this line of work, but asks a more focused methodological question: whether replacing interpolation by a nonlinear inverse model can improve rainfall reconstruction while also reproducing the observed link attenuations more faithfully.

## Chapter 3

# Methodology

### 3.1 Data and Benchmark Construction

Our benchmark is built patch by patch. A *patch* is a rectangular crop of an hourly rainfall accumulation map from the OPERA European weather-radar product, selected to capture a detected rain event, with the patch width and height constrained to lie between 50 km and 95 km. For each patch in the benchmark list, we first extract the corresponding rainfall crop from the OPERA map. We then refine that crop from the native 2 km grid to a finer 125 m grid by splitting each coarse OPERA pixel into smaller subpixels and assigning each subpixel the parent pixel’s rainfall value. We then smooth this refined field. This smoothed high-resolution rainfall field serves as ground truth for both simulating link attenuations and evaluating the solvers. [Thus, although the link attenuations are simulated, the underlying rainfall fields are based on observed OPERA rain events rather than randomly generated synthetic patterns; the refinement and smoothing steps provide a high-resolution field while preserving the large-scale structure of the radar-derived event.](#) The microwave-link data set, obtained from the 4TU-NL link data, is then overlaid on each patch, and only links whose endpoints fall inside the patch are retained. Each retained link preserves its original metadata (endpoints and frequency) from the data set and is included in the benchmark only if both endpoints lie within the selected patch. The 4TU-derived link file used in this benchmark does not specify polarization, so the patch-export pipeline supplies a default polarization when missing. In the present benchmark, that default is set to horizontal (H), and the forward model therefore uses the ITU coefficients corresponding to horizontal polarization. Because the patch detector is designed to capture rainy events rather than dry background, the benchmark is rain-dominated: across the 100 patches, the mean non-rainy proportion is about 10.50 %, where non-rainy pixels are those with very little or no rainfall.

This benchmark design differs from much of the earlier literature in a useful way. Some influential studies work directly with operational microwave-link measurements and validate against radar and gauges (Messer et al., 2006; Roversi et al., 2020), while others study interpolation behavior on synthetic rain fields and synthetic CML networks (Eshel et al., 2020). Here we instead begin from a radar-derived rainfall field that is treated as known ground truth, synthesize the corresponding link attenuations under a shared forward model, and then compare the inverse solvers under that controlled setup. This gives us exact access to both the rainfall field and the attenuation field for every test case, which makes solver-to-solver comparisons cleaner. It also yields a relatively large patch-wise reconstruction benchmark: the present data set contains 100 patches, with between 939 and 1996 links per patch (mean 1324), and the largest-area patch contains 1894 links. For comparison, Roversi et al. (2020) evaluate their operational study on 357 links in northern Italy. The comparison is not one-to-one, but it helps place the scale of the present benchmark.

#### 3.1.1 Geographic Distribution of the Benchmark Patches

Figure 1 maps the geographic distribution of the 100 benchmark patches. Each red rectangle marks the footprint of one selected patch on the Europe basemap. The figure makes clear that the benchmark is not concentrated in a single country or climatological regime: the selected rainy-event patches are spread across northern, western, central, southern, and southeastern Europe.



Figure 1: Geographic distribution of the 100 benchmark patches. Each patch is shown as a red outline rectangle over a Europe basemap. The figure visualizes the spatial spread of the selected rainy-event patches across the OPERA domain and provides geographic context for the benchmark used throughout the thesis.

### 3.1.2 Patch-Geometry Overview

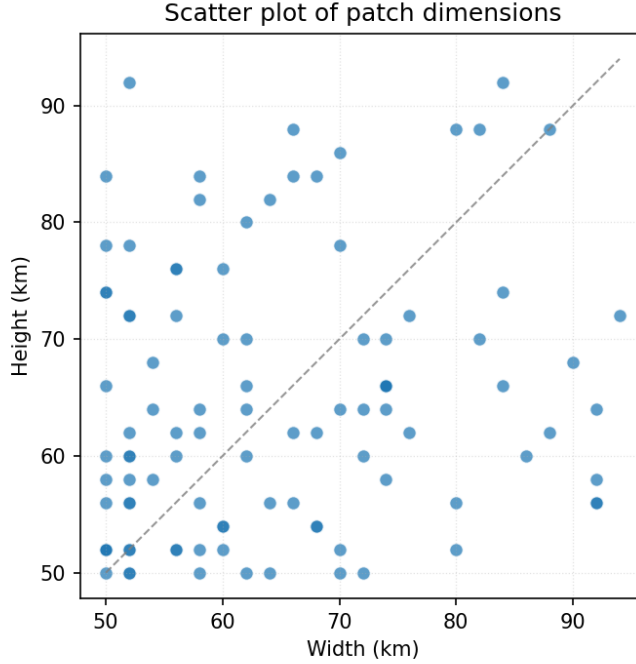


Figure 2: Patch width versus patch height for the 100 benchmark patches, with one dot per patch. Across patches, the mean patch area is 269,491 pixels with SD of 76,755 pixels (pixels are squares with side length 125 m).

| Patch size (km) | # links | 1-multiplicity | 2-multiplicity |
|-----------------|---------|----------------|----------------|
| $50 \times 50$  | 939     | 99             | 420            |
| $62 \times 60$  | 1254    | 128            | 563            |
| $72 \times 70$  | 1507    | 157            | 675            |

Table 1: Representative link multiplicities for three patch dimensions. The multiplicity of a link is defined as the number of frequencies assigned to it. The second column reports the total number of link–frequency pairs. In the data set, link multiplicities were either 1 or 2, with approximately 10% of links having multiplicity 1.

## 3.2 Forward Model

In the benchmark, link-level observations (attenuation and link metadata) are generated from the known ground-truth rainfall field via a forward model. Starting from the ground-truth rainfall map, each link is represented as a line segment between its endpoints, and attenuation is computed using ITU-R P.838-3, the standard relation for rain-induced specific attenuation in microwave links. In this model, the specific attenuation (dB/km) is

$$\gamma_R = \kappa(f, \text{pol}) R^{\alpha(f, \text{pol})},$$

where  $R$  is rain rate (mm/h), and  $(\kappa, \alpha)$  depend on link frequency and polarization. The total link attenuation is then the line integral of specific attenuation along the propagation path,

$$A_\ell \triangleq \int_\ell \gamma_R(R(s)) \, ds.$$

In the benchmark, and likewise for the methods we compare against, this integral is evaluated in discretized form on the refined 125 m grid: each link is traced through crossed grid cells, and attenuation is accumulated as a length-weighted sum of cellwise specific attenuations,

$$A_\ell \approx \sum_{c \in \mathcal{C}(\ell)} \kappa_\ell R_c^{\alpha_\ell} \Delta s_{\ell,c},$$

where  $\mathcal{C}(\ell)$  is the set of crossed cells and  $\Delta s_{\ell,c}$  is the path length of link  $\ell$  inside cell  $c$  (in km). These computed attenuation values, together with link metadata, form the link-level observations used for inverse reconstruction of the rainfall field.

### 3.3 Inverse Solvers

We compare IDW, ILDW, and four optimization-based reconstructions: Solver(ILDW), Convex Solver, Homotopy Solver, and Solver(GT). Solver(GT) is initialized from the ground-truth rainfall field, so it is not a method that could be used on real unknown rainfall fields. Instead, we use it as an empirical best-case reference for this objective and optimizer.

Using the same link-level observations and forward-model conventions described above, we now describe each solver family and its implementation details.

#### 3.3.1 IDW

Inverse-distance weighting (IDW) is one of the standard simple interpolation strategies used in commercial-microwave-link rainfall reconstruction (Chwala and Kunstmann, 2019). The input consists of a set of links, where each link carries total attenuation, frequency, and polarization.

IDW first converts each observed link attenuation  $A_\ell$  into a single link-level rain-rate estimate defined as:

$$R_\ell \triangleq \left( \frac{A_\ell / |\ell|}{\kappa_\ell} \right)^{1/\alpha_\ell}.$$

where  $A_\ell$  denotes the attenuation of link  $\ell$ ,  $|\ell|$  its length, and  $(\kappa_\ell, \alpha_\ell)$  the ITU-R P.838-3 coefficients corresponding to its frequency and polarization.

Each per-link rain-rate estimate is then treated as a virtual point observation at the link midpoint. For each pixel  $p$  with center  $c_p$ , IDW computes a weighted average of nearby  $R_\ell$  values, using weights proportional to  $\|m_\ell - c_p\|_2^{-2}$  within a prescribed search radius  $r_{\max}$  (set to 15km in our study). If one or more link midpoints lie inside pixel  $p$ ,  $\hat{R}_p$  (the estimated rainfall in pixel  $p$ ), is set to the mean of their rainfall estimates. This defines the discretized IDW baseline used throughout this thesis.

---

**Algorithm 1:** Discretized IDW. Each link is replaced with a virtual gauge located at the midpoint of the link. The rain rate at the gauge is set to the uniform rain rate that reproduces the input link attenuation under ITU-R P.838-3. IDW is then applied with respect to these virtual gauges. We set the exponent of the decay to 2.

---

**Data:** link set  $\mathcal{L}$ , a patch  $P$ , and a maximum influence distance  $r_{\max}$

**Result:** rainfall field  $\hat{R}$

```

1 foreach link  $\ell \in \mathcal{L}$  do                                     // compute rain rate per link
2    $(\ell.\kappa, \ell.\alpha) \leftarrow \text{ITU838}(\ell.\text{freq}, \ell.\text{polarization})$ 
3    $\ell.\text{rainrate} \leftarrow \left( \frac{\ell.\text{attenuation}/\ell.\text{length}}{\ell.\kappa} \right)^{1/\ell.\alpha}$ 
4 foreach pixel  $p \in P$  do                                       // compute rain  $\hat{R}_p$ 
5    $M_p \leftarrow \{\ell \in \mathcal{L} : \ell.\text{midpoint} \in p\}$ 
6   if  $M_p \neq \emptyset$  then
7      $\hat{R}_p \leftarrow \frac{1}{|M_p|} \sum_{\ell \in M_p} \ell.\text{rainrate}$            // pixel contains a link endpoint
8   else
9      $N_p \leftarrow \{\ell \in \mathcal{L} : \|\ell.\text{midpoint} - p.\text{center}\|_2 \leq r_{\max}\}$ 
10    if  $N_p = \emptyset$  then
11       $\hat{R}_p \leftarrow 0$                                            // pixel far from links
12    else
13       $\hat{R}_p \leftarrow \sum_{\ell \in N_p} \frac{\|\ell.\text{midpoint} - p.\text{center}\|_2^{-2}}{\sum_{\ell' \in N_p} \|\ell'.\text{midpoint} - p.\text{center}\|_2^{-2}} \cdot \ell.\text{rainrate}$ 

```

---

### 3.3.2 ILDW

ILDW is a link-aware modification of IDW used in this work. It starts from the same per-link rain-rate values obtained by inverting the discretized attenuation forward model for each link. The motivation for this variant is that a microwave-link attenuation measurement is associated with an entire path, not only with the midpoint of that path. We therefore considered whether interpolation could be improved by using the full link geometry when deciding which links should influence a pixel and how strongly they should do so.

More precisely, for a pixel center  $c_p$ , ILDW replaces the midpoint distance  $\|m_\ell - c_p\|_2$  by the point-to-segment distance

$$\text{dist}(c_p, \ell),$$

where  $\ell = [p_{\ell,1}, p_{\ell,2}]$  denotes the full link segment with endpoint coordinates  $p_{\ell,1}$  and  $p_{\ell,2}$ .

If one or more links cross the pixel, the estimate is taken to be the mean of the corresponding link rain-rate values. Otherwise, the estimate is a weighted average over nearby links within radius  $r_{\max}$  (set to 15 km), with weights proportional to the inverse squared segment distance. We considered this variant important because it preserves the same simple interpolation philosophy as IDW while using more of the link-path geometry.



---

**Algorithm 2:** Discretized ILDW. Rainfall is averaged by inverse squared distance to a link. Lines that are different from their corresponding lines in IDW are colored red.

---

**Data:** link set  $\mathcal{L}$ , a patch  $P$ , and a maximum influence distance  $r_{\max}$   
**Result:** rainfall field  $\hat{R}$

```

1 foreach link  $\ell \in \mathcal{L}$  do                                     // compute rain rate per link
2    $(\ell.\kappa, \ell.\alpha) \leftarrow \text{ITU838}(\ell.\text{freq}, \ell.\text{polarization})$ 
3    $\ell.\text{rainrate} \leftarrow \left( \frac{\ell.\text{attenuation}/\ell.\text{length}}{\ell.\kappa} \right)^{1/\ell.\alpha}$ 
4 foreach pixel  $p \in P$  do                                     // compute rain  $\hat{R}_p$ 
5    $X_p \leftarrow \{\ell \in \mathcal{L} : \ell \cap p \neq \emptyset\}$ 
6   if  $X_p \neq \emptyset$  then
7      $\hat{R}_p \leftarrow \frac{1}{|X_p|} \sum_{\ell \in X_p} \ell.\text{rainrate}$            // at least one link traverses the pixel
8   else
9      $N_p \leftarrow \{\ell \in \mathcal{L} : \text{dist}(p, \ell) \leq r_{\max}\}$ 
10    if  $N_p = \emptyset$  then
11       $\hat{R}_p \leftarrow 0$                                            // pixel far from links
12    else
13       $\hat{R}_p \leftarrow \sum_{\ell \in N_p} \frac{\text{dist}^{-2}(p.\text{center}, \ell)}{\sum_{\ell' \in N_p} \text{dist}^{-2}(p.\text{center}, \ell')} \cdot \ell.\text{rainrate}$ 

```

---

### 3.3.3 Optimization-Based Reconstruction

The optimization-based reconstructions explore a different approach from purely interpolation-based rainfall mapping. IDW and ILDW first reduce each link to a link-level rain estimate and then spread that information spatially by geometric weighting. In contrast, the optimization-based methods treat the rainfall value in every pixel as an unknown and choose the full rainfall field by minimizing an objective that combines agreement with the observed link attenuations with terms that encourage spatially coherent solutions.

**Optimization Objective** For each patch  $P$ , we identify  $P$  with its set of pixels, and denote by  $\mathcal{L}(P)$  the set of links contained in  $P$ . Let  $r \in \mathbb{R}_{\geq 0}^{|P|}$  denote the unknown rainfall field written as a vector over pixels. We solve the constrained optimization problem

$$\min_{r \in \mathbb{R}_{\geq 0}^{|P|}} J(r),$$

where the objective is a weighted sum of four terms:

$$J(r) \triangleq \alpha_{\text{atten}} J_{\text{atten}}(r) + \alpha_{1d} J_{1d}(r) + \alpha_{2d} J_{2d}(r) + \alpha_{\text{total}} J_{\text{total}}(r).$$

Here  $J_{\text{atten}}$  penalizes disagreement between observed and forward-model-predicted link attenuations, while  $J_{1d}$ ,  $J_{2d}$ , and  $J_{\text{total}}$  are regularization terms that encourage the estimated field to remain spatially and physically well behaved. In the normalized solver studied here, we define the

weights instance by instance using the corresponding objective values at the ILDW solution:

$$\begin{aligned}\alpha_{\text{atten}} &\triangleq \frac{1}{J_{\text{atten}}(\text{ILDW})}, & \alpha_{1d} &\triangleq \frac{1}{J_{1d}(\text{ILDW})}, \\ \alpha_{2d} &\triangleq \frac{1}{2} \cdot \frac{1}{J_{2d}(\text{ILDW})}, & \alpha_{\text{total}} &\triangleq \frac{1}{|P|} \cdot \frac{1}{J_{\text{total}}(\text{ILDW})}.\end{aligned}$$

More concretely,  $J_{\text{atten}}$  is the attenuation-misfit term:

$$J_{\text{atten}}(r) \triangleq \frac{1}{|\mathcal{L}(P)|} \sum_{\ell \in \mathcal{L}(P)} \frac{(\hat{A}_\ell(r) - A_\ell^{\text{obs}})^2}{|\ell|},$$

where  $\hat{A}_\ell(r)$  is the attenuation predicted by the forward model from the rainfall field  $r$ ,  $A_\ell^{\text{obs}}$  is the observed attenuation of link  $\ell$ , and  $|\ell|$  is the link length in km. Thus  $J_{\text{atten}}$  measures how far the forward-model-implied attenuations are from the observed attenuations, with a normalization by link length and averaging over the links in the patch.

The term  $J_{1d}$  is a first-difference penalty:

$$J_{1d}(r) \triangleq \frac{1}{|P|} \sum_{(u,v) \in \mathcal{E}(P)} (r_u - r_v)^2,$$

where  $\mathcal{E}(P)$  is the set of horizontally and vertically adjacent pixel pairs in patch  $P$ . This term sums squared differences between neighboring pixel values and therefore discourages abrupt pixel-to-pixel jumps in the estimated rainfall field.

The term  $J_{2d}$  is a second-difference penalty:

$$J_{2d}(r) \triangleq \frac{1}{|P|} \sum_{(p,q,t) \in \mathcal{T}(P)} ((r_q - r_p) - (r_t - r_q))^2,$$

where  $\mathcal{T}(P)$  is the set of horizontal and vertical triples of collinear pixels in patch  $P$ . It penalizes changes in the first difference, so it discourages unnecessary curvature or oscillation along rows and columns.

Finally,  $J_{\text{total}}$  is an  $\ell_2$  shrinkage term on the rainfall field itself:

$$J_{\text{total}}(r) \triangleq \frac{1}{|P|} \sum_{p \in P} r_p^2.$$

This term biases the solution toward smaller overall rainfall values, except where larger values are needed to match the observed link attenuations.

These four terms are added together only after weighting, so the optimizer balances attenuation fit against smoothness, curvature control, and shrinkage.

**Solver Variants** All optimization-based variants are solved iteratively with L-BFGS-B, a limited-memory quasi-Newton method for smooth constrained optimization (Byrd et al., 1995; Zhu et al., 1997). Starting from an initial rainfall field, the optimizer repeatedly updates the field to reduce the objective while enforcing the nonnegativity bounds on rainfall. In the present setting, the optimization variable is the rainfall value at every pixel of the patch. We write

$$\hat{r} \leftarrow \text{L-BFGS-B}(\mathcal{L}, P, r_{\text{init}})$$

to denote one invocation of L-BFGS-B on patch  $P$  with link set  $\mathcal{L}$  and initial rainfall field  $r_{\text{init}}$ .

**Solver(ILDW)** The first variant applies L-BFGS-B directly to the original nonconvex objective above, using the ILDW rainfall field as the initial point:

$$\hat{r}_{\text{ILDW}} \leftarrow \text{L-BFGS-B}(\mathcal{L}, P, r_{\text{ILDW}}).$$

The objective is generally nonconvex because the ITU-R attenuation model uses a power law in rainfall. For a link  $\ell$ , the predicted attenuation has the form

$$\hat{A}_\ell(r) = \sum_{c \in \mathcal{C}(\ell)} \kappa_\ell r_c^{\alpha_\ell} \Delta s_{\ell,c}.$$

When  $\alpha_\ell = 1$ , this predicted attenuation is linear in the pixel rain rates, so the squared attenuation residual is convex. When  $\alpha_\ell \neq 1$ , the squared residual is not generally convex; even in one dimension,  $(r^\alpha - A)^2$  can have negative curvature for positive observed attenuation  $A$ . Thus the solution found by L-BFGS-B can depend on the initial field.

We use ILDW as the initial point for this nonconvex solve because it is already constructed to be consistent with individual link attenuations in the simple case where links do not overlap. If links were pairwise disjoint at the pixel level, assigning each crossed pixel the rain-rate estimate of the link that crosses it would reproduce each observed attenuation exactly under the same forward model. Real patches contain overlapping links, so this exact consistency cannot generally hold for all links at once, but ILDW remains a natural initialization because it is organized around the same link paths whose attenuations the optimizer tries to fit.

**Convex Solver** The second variant constructs a convex surrogate inverse problem by replacing each original link with a virtual link. The virtual link keeps the same geometry and length, but uses a reference frequency  $f^0$  for which the ITU-R attenuation exponent is one. In the present benchmark the links are treated as horizontally polarized, for which this reference frequency is  $f^0 = 24.92$  GHz. Let  $R_\ell$  denote the link-level rain-rate equivalent of the original observed attenuation. The virtual observed attenuation is then

$$A_\ell^0 \triangleq |\ell| \kappa_0 R_\ell,$$

where  $\kappa_0$  is the ITU-R coefficient at  $f^0$ . We denote the resulting virtual link set by  $\mathcal{L}^0$ . Because the attenuation exponent is one for these virtual links, the predicted attenuation is linear in the rainfall field, so the attenuation-misfit term is a sum of squared linear residuals. Together with the convex regularization terms, this gives a convex surrogate objective. We write

$$\hat{r}^0 \leftarrow \text{L-BFGS-B}(\mathcal{L}^0, P, r_{\text{init}})$$

for this solve. In the final experiments, this is the method reported as Convex Solver.

**Homotopy Solver** The third variant uses homotopy continuation (Allgower and Georg, 2003) to move gradually from the convex surrogate problem back toward the original nonconvex formulation. For  $\lambda \in [0, 1]$ , define a virtual link set  $\mathcal{L}^\lambda$  in which each link keeps the same geometry, length, and polarization as the original link, but has interpolated frequency

$$f_\ell^\lambda \triangleq (1 - \lambda) f^0 + \lambda f_\ell.$$

Let  $\kappa_\ell(\lambda)$  and  $\alpha_\ell(\lambda)$  be the ITU-R coefficients at frequency  $f_\ell^\lambda$ . The corresponding virtual observed attenuation is

$$A_\ell^\lambda \triangleq |\ell| \kappa_\ell(\lambda) R_\ell^{\alpha_\ell(\lambda)}.$$

Starting from the convex problem at  $\lambda = 0$ , the solver advances through a sequence of increasing  $\lambda$  values, each time initializing from the previous solution:

$$\hat{r}^0 \leftarrow \text{L-BFGS-B}(\mathcal{L}^0, P, r_{\text{init}}), \quad \hat{r}^\lambda \leftarrow \text{L-BFGS-B}(\mathcal{L}^\lambda, P, \hat{r}^{\lambda-\delta}).$$

The final output is  $\hat{r}^1$ , which corresponds to the original link frequencies. In the final experiments, this continuation-based method is reported as Homotopy Solver, and we use a continuation step size of  $\delta = 0.1$ .

**Solver(GT)** We also include a ground-truth-initialized diagnostic run:

$$\hat{r}_{\text{GT}} \leftarrow \text{L-BFGS-B}(\mathcal{L}, P, r_{\text{GT}}).$$

This run uses information unavailable in practice, so it is not a deployable reconstruction method. Instead, it provides an optimistic empirical reference for interpreting the objective landscape and for seeing what the objective preserves, or moves away from, when initialized at the target rainfall field.

## Chapter 4

# Evaluation

### 4.1 Evaluation Protocol

We evaluate each solver in four ways: overall agreement with the ground-truth rainfall map, how errors change as distance from links increases, wet/dry classification behavior, and agreement with observed link attenuations under the forward model. We first report patch-level summaries across all 100 benchmark patches, and then provide qualitative case studies in the appendix. All optimization-based results reported below use the same objective terms and weights described above, with a maximum budget of 2000 optimizer iterations per patch.

#### 4.1.1 Optimization Stopping Criteria

The 2000-iteration budget was used as a cap rather than as a fixed number of iterations. Each optimized patch reconstruction used the same L-BFGS-B stopping tolerances, with relative objective-reduction tolerance  $\text{ftol} = 10^{-6}$  and projected-gradient tolerance  $\text{gtol} = 10^{-6}$ . Let  $J_k$  denote the objective value at iteration  $k$ . The  $\text{ftol}$  criterion stops the solve when the relative objective decrease between two successive iterations is small,

$$\frac{J_k - J_{k+1}}{\max\{|J_k|, |J_{k+1}|, 1\}} \leq \text{ftol}.$$

The  $\text{gtol}$  criterion stops the solve when the projected gradient is small,

$$\|g_{k+1}^{\text{proj}}\|_{\infty} \leq \text{gtol},$$

where  $g_{k+1}^{\text{proj}}$  is the gradient projected according to the bound constraints used by L-BFGS-B. Intuitively, the first criterion means that the objective is no longer decreasing meaningfully, while the second means that the bound-constrained optimizer is close to a stationary point. These tolerances were chosen to require clear numerical convergence without forcing the optimizer to spend many iterations chasing negligible objective changes. This is important for interpreting the results: the reported behavior is not explained by a shortage of iterations. Across the 100 patches and four optimization-based solvers, none of the 400 optimized runs reached the 2000-iteration cap. Instead, the runs stopped once the configured convergence tolerances were met.

| Solver          | Mean iter. | SD iter. | Min | Max  | Stop reason     |
|-----------------|------------|----------|-----|------|-----------------|
| Solver(ILDW)    | 557.4      | 289.1    | 3   | 1198 | 100 ftol        |
| Convex Solver   | 936.7      | 233.8    | 387 | 1557 | 99 ftol, 1 gtol |
| Homotopy Solver | 1074.6     | 243.0    | 430 | 1715 | 100 ftol        |
| Solver(GT)      | 594.4      | 345.9    | 3   | 1787 | 100 ftol        |

Table 2: Iteration counts for the optimization-based reconstruction runs across the 100 evaluated patches. The standard deviation is computed across patches. No optimized run reached the 2000-iteration cap; all runs stopped because either the relative objective-reduction tolerance  $\text{ftol} = 10^{-6}$  or the projected-gradient tolerance  $\text{gtol} = 10^{-6}$  was met.

**Key findings at a glance.** Across the benchmark, the optimized solvers improve substantially over IDW, with Convex Solver and Homotopy Solver giving stronger rainfall-map agreement than Solver(ILDW) while also preserving strong attenuation consistency. Homotopy Solver does not improve the map-level RMSE, bias, or correlation summaries relative to Convex Solver, although it does provide a small gain in attenuation consistency. Solver(GT) gives the best map agreement, but Convex Solver and Homotopy Solver achieve lower attenuation mismatch, showing that pixelwise agreement with the radar-derived ground truth and consistency with the link observations are related but not identical criteria. When we look at distance bins, all methods degrade as nearby-link coverage becomes weaker, but the optimized solvers remain clearly better than IDW in the low-coverage bins.

## 4.2 Patch-Level Metrics

We now turn to the empirical comparison on the hundred-patch benchmark. We begin with whole-patch summaries, then examine how errors vary with link proximity, and finally report attenuation-consistency analyses.

For a patch  $P$ , let  $R_P(p)$  denote the ground-truth rainfall at pixel  $p$ , and let  $\hat{R}_{P,s}(p)$  denote the rainfall predicted at pixel  $p$  by solver  $s$ . We compute three scalar metrics over all pixels in  $P$ : the root mean squared error

$$\text{RMSE}_{P,s} \triangleq \sqrt{\frac{1}{|P|} \sum_{p \in P} (\hat{R}_{P,s}(p) - R_P(p))^2},$$

the signed bias

$$\text{Bias}_{P,s} \triangleq \frac{1}{|P|} \sum_{p \in P} (\hat{R}_{P,s}(p) - R_P(p)),$$

and the Pearson correlation coefficient

$$r_{P,s} \triangleq \frac{\sum_{p \in P} (\hat{R}_{P,s}(p) - \bar{\hat{R}}_{P,s})(R_P(p) - \bar{R}_P)}{\sqrt{\sum_{p \in P} (\hat{R}_{P,s}(p) - \bar{\hat{R}}_{P,s})^2} \sqrt{\sum_{p \in P} (R_P(p) - \bar{R}_P)^2}},$$

where

$$\bar{\hat{R}}_{P,s} \triangleq \frac{1}{|P|} \sum_{p \in P} \hat{R}_{P,s}(p), \quad \bar{R}_P \triangleq \frac{1}{|P|} \sum_{p \in P} R_P(p).$$

RMSE measures overall magnitude error. Bias measures average signed error (net over- or underestimation). Correlation measures how well the spatial structure of the field is reproduced independently of scale and offset. The Pearson correlation coefficient satisfies  $r_{P,s} \in [-1, 1]$ . A value of 1 indicates perfect positive linear agreement between predicted and true rainfall maps. A value of 0 indicates no linear relationship. A value of  $-1$  indicates perfect negative linear agreement (an exact inverse pattern).

| Solver          | RMSE              | Bias              | Correlation       |
|-----------------|-------------------|-------------------|-------------------|
| IDW             | $1.307 \pm 1.234$ | $0.034 \pm 0.296$ | $0.866 \pm 0.059$ |
| ILDW            | $1.239 \pm 1.182$ | $0.047 \pm 0.261$ | $0.878 \pm 0.055$ |
| Solver(ILDW)    | $1.059 \pm 1.192$ | $0.040 \pm 0.235$ | $0.918 \pm 0.055$ |
| Convex Solver   | $0.845 \pm 0.803$ | $0.023 \pm 0.075$ | $0.940 \pm 0.034$ |
| Homotopy Solver | $0.845 \pm 0.803$ | $0.023 \pm 0.075$ | $0.940 \pm 0.034$ |
| Solver(GT)      | $0.503 \pm 0.369$ | $0.009 \pm 0.030$ | $0.961 \pm 0.032$ |

Table 3: Descriptive patch-level summary of three global map metrics across the 100 evaluated patches. Each entry reports mean  $\pm$  standard deviation across patches. Here RMSE and Bias are computed in the same rainfall units as  $R_P$  and  $\hat{R}_{P,s}$ , while Correlation is the Pearson correlation coefficient between the predicted and ground-truth rainfall fields within each patch.

Table 3 gives a descriptive overview of the scale and variability of the patch-level metrics for each solver. Homotopy Solver gives nearly identical descriptive metrics to Convex Solver, so in these experiments its added continuation machinery does not produce an additional empirical gain. Solver(GT) achieves the strongest descriptive agreement with the ground-truth rainfall maps, as expected for a run initialized from the ground truth. Comparing IDW and ILDW also suggests that keeping more of the link-path geometry can be useful even within a simple interpolation framework: ILDW improves the mean RMSE and correlation relative to midpoint IDW, although its mean signed bias is not lower than IDW’s.

Because all solvers are evaluated on the same 100 patches, direct solver comparisons should also use the paired patchwise differences. Here we restrict the paired comparison to Convex Solver against IDW, since this is the main optimized solver relative to the standard interpolation baseline. In that paired view, Convex Solver improves strongly over IDW: it has lower RMSE on all 100 patches and higher Pearson correlation on all 100 patches. For absolute bias, where the result is not a clean sweep, Convex Solver is closer to zero bias on 78 patches and IDW is closer on 22 patches.

For absolute bias, we compare the paired patchwise differences

$$d_P = |\text{Bias}_{P,\text{Convex}}| - |\text{Bias}_{P,\text{IDW}}|.$$

Negative values mean that Convex Solver is closer to zero bias on patch  $P$ . We use a one-sided paired mean test of  $H_0 : \mathbb{E}[d_P] \geq 0$  against  $H_1 : \mathbb{E}[d_P] < 0$ ; this test rejects the null at  $\alpha = 0.01$ .

| Metric              | Better direction | Convex wins | IDW wins | Paired evidence                          |
|---------------------|------------------|-------------|----------|------------------------------------------|
| RMSE                | lower            | 100         | 0        | all patches favor Convex Solver          |
| Absolute Bias       | lower            | 78          | 22       | rejects null: $p = 4.667 \times 10^{-7}$ |
| Pearson correlation | higher           | 100         | 0        | all patches favor Convex Solver          |

Table 4: Paired patchwise comparison between Convex Solver and IDW across the 100 evaluated patches.

### 4.2.1 Attenuation Mismatch

The previous metrics compare each reconstructed rainfall map directly to the ground-truth radar field. We also evaluate the reconstructions from the point of view of the CML observations themselves: if a predicted rainfall field is passed through the forward attenuation model, how closely

does it reproduce the measured link attenuations? This matters because a useful reconstruction should not only resemble the radar field pixelwise, but should also remain consistent with the link measurements that drive the inverse problem.

We report two complementary attenuation-mismatch summaries in the same table. The mean absolute attenuation error per kilometer gives a directly interpretable link-level discrepancy in dB/km, while  $J_{\text{atten}}$  reports the squared attenuation-fitting term used inside the optimization objective.

| Solver          | Abs. err. mean<br>(dB/km) | Abs. err. SD<br>(dB/km) | $J_{\text{atten}}$ mean<br>(dB <sup>2</sup> /km) | $J_{\text{atten}}$ SD<br>(dB <sup>2</sup> /km) |
|-----------------|---------------------------|-------------------------|--------------------------------------------------|------------------------------------------------|
| IDW             | 0.05034                   | 0.03396                 | 0.01522                                          | 0.03808                                        |
| ILDW            | 0.01356                   | 0.00943                 | 0.00289                                          | 0.00757                                        |
| Solver(ILDW)    | 0.00207                   | 0.00373                 | 0.00018                                          | 0.00081                                        |
| Convex Solver   | 0.00107                   | 0.00099                 | 0.000014                                         | 0.000050                                       |
| Homotopy Solver | 0.00074                   | 0.00067                 | 0.0000065                                        | 0.000024                                       |
| Solver(GT)      | 0.00129                   | 0.00166                 | 0.000063                                         | 0.000217                                       |

Table 5: Patch-level attenuation-mismatch summaries across the 100 evaluated patches. For a given patch and solver  $s$ , the absolute-error columns summarize  $\frac{1}{N_{\text{links}}} \sum_{\ell \in \text{links}} \frac{|\hat{A}_{\ell,s} - A_{\ell}^{\text{obs}}|}{|\ell|}$ , while  $J_{\text{atten}} = \frac{1}{N_{\text{links}}} \sum_{\ell \in \text{links}} \frac{(\hat{A}_{\ell,s} - A_{\ell}^{\text{obs}})^2}{|\ell|}$ . Here  $\hat{A}_{\ell,s}$  is the attenuation implied on link  $\ell$  by the rainfall field estimated by solver  $s$ ,  $A_{\ell}^{\text{obs}}$  is the observed attenuation, and  $|\ell|$  is the link length in kilometers. Lower values indicate better agreement with the observed link attenuations.

Together, these attenuation-based summaries show that the optimized solvers match the measured link attenuations much more closely than IDW and ILDW, with Convex Solver and Homotopy Solver giving the smallest mismatches. They also add a useful qualification to the map-level results above: pixelwise agreement with the radar-derived ground truth and consistency with the CML observations are related but not identical criteria: Solver(GT) remains closest to the radar field, but Convex Solver and Homotopy Solver better satisfy the link observations under the forward model. Thus, the field closest to the radar ground truth is not necessarily the field that minimizes attenuation mismatch. In addition, although Homotopy Solver does not improve the RMSE, bias, or correlation summaries relative to Convex Solver, it does provide a small gain in link consistency.

## 4.3 Distance-Conditioned Error Analysis

We next ask how prediction quality varies with link proximity. To answer this question, we group pixels into distance bins according to their distance to the third-closest link and summarize patch-level error statistics within each bin. We first examine rainy pixels through relative absolute error, and then examine non-rainy pixels through absolute error. In this subsection we focus exclusively on distance-conditioned pixelwise error summaries.

### 4.3.1 Rainy Pixel Relative Absolute Error by Link Distance

We define a pixel  $p$  to be rainy if  $R_P(p) \geq 0.6$  mm/h, and we write

$$P_{\text{rain}} = \{p \in P : R_P(p) \geq 0.6\}.$$



For a rainy pixel  $p$ , the relative absolute error with respect to solver  $s$  is

$$\text{RAE}_{P,s}(p) \triangleq \frac{|R_P(p) - \hat{R}_{P,s}(p)|}{R_P(p)}.$$

To study how prediction quality changes with link proximity, we group pixels by distance to the third-closest link. For a pixel  $p \in P$ , let  $c(p) \in \mathbb{R}^2$  denote the center of pixel  $p$ . For a link  $\ell \in \mathcal{L}(P)$ , viewed geometrically as a line segment, let

$$\text{dist}(p, \ell) \triangleq \text{dist}(c(p), \ell)$$

denote the Euclidean distance from the pixel center to that link segment. If the links in patch  $P$  are written as  $\ell_1, \dots, \ell_{|\mathcal{L}(P)|}$ , we define

$$d_3(p) \triangleq \text{the third-order statistic of } \{\text{dist}(p, \ell_j)\}_{j=1}^{|\mathcal{L}(P)|},$$

that is, the distance from the center of pixel  $p$  to its third-closest link segment.

For each index  $i$ , let  $b_{i-1}$  and  $b_i$  denote the lower and upper distance cutoffs of the  $i$ th bin, and write  $B_i = (b_{i-1}, b_i]$ . A pixel  $p$  belongs to bin  $B_i$  exactly when  $d_3(p) \in B_i$ .

For each patch  $P$ , solver  $s$ , and distance bin  $B_i$ , we then collect the rainy-pixel RAE values in that bin and summarize them by the patch-level median

$$m_{s,i}^{\text{rain}}(P) \triangleq \text{median}\{\text{RAE}_{P,s}(p) : p \in P_{\text{rain}}, d_3(p) \in B_i\}.$$

In Figure 3, these patch-level medians are compared across the full benchmark collection  $\mathcal{P}$  of 100 patches. There, the displayed distance bins are

$$[0, 125] \text{ m}, (125, 375] \text{ m}, (375, 750] \text{ m}, (750, 1500] \text{ m}, (1500, 3125] \text{ m}, (3125, 6000] \text{ m}, (6000, \infty) \text{ m},$$

In the box-and-whisker plot, the two sparsely populated tail bins  $(6000, 9000] \text{ m}$  and  $(9000, \infty)$  are merged and displayed as a single  $(6000, \infty) \text{ m}$  bin. The count table below still reports these two tail bins separately.

For fixed solver  $s$  and distance bin  $B_i$ , let

$$\mu_{s,i}^{\text{rain}} \triangleq \{m_{s,i}^{\text{rain}}(P)\}_{P \in \mathcal{P}}$$

denote the sequence of patch-level rainy-pixel median RAE values across patches. Each box-and-whisker then summarizes this sequence: the lower whisker tip is the minimum of  $\mu_{s,i}^{\text{rain}}$ , the lower edge of the box is its 25th percentile, the horizontal line segment with the filled circle at its center marks the median, the upper edge of the box is its 75th percentile, and the upper whisker tip is its maximum.

### 4.3.2 Non-Rainy Pixel Absolute Error by Link Distance

For the dry-pixel analysis, we focus on pixels whose ground-truth rainfall falls below the wet threshold. We therefore define the non-rainy pixel set by

$$P_{\text{nonrain}} = \{p \in P : R_P(p) < 0.6\}.$$

For these pixels, relative normalization by  $R_P(p)$  is not appropriate because the denominator may be zero or very small. We therefore measure prediction quality by absolute error rather than relative absolute error:

$$\text{AE}_{P,s}(p) = |R_P(p) - \hat{R}_{P,s}(p)|.$$

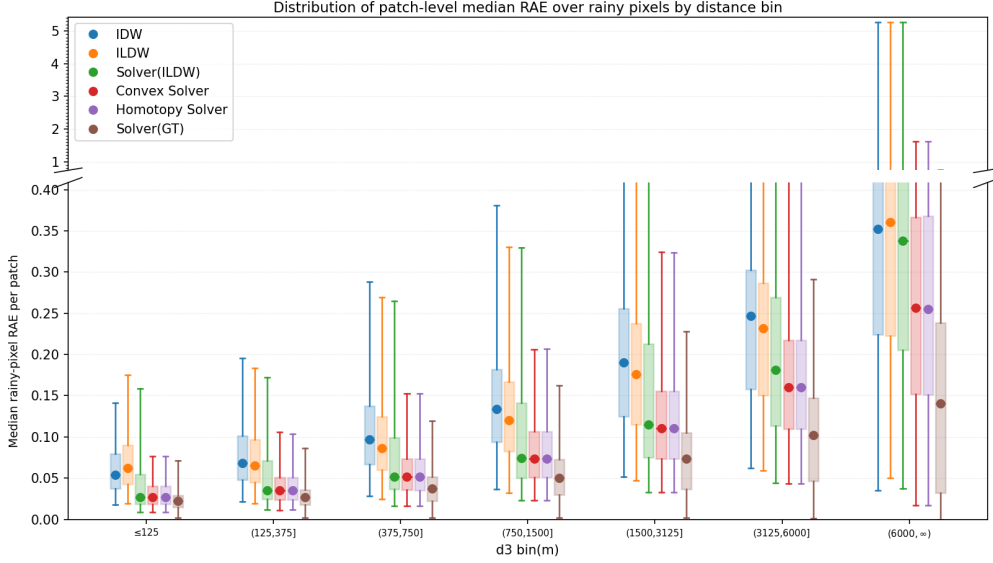


Figure 3: Rainy-pixel distance-profile box-and-whisker summary for distance to the third closest link. Each patch contributes one median rainy-pixel relative-absolute-error (RAE) value per distance bin, and these patch-level medians are then summarized across patches with a box-and-whisker plot. Rainy pixels are those whose ground-truth rainfall is at least the configured wet threshold, set here to 0.6 mm/h. The corresponding rainy-pixel counts are reported separately in Table 6.

| Distance to third closest link (m) | Rainy mean count (% of total) | SD       |
|------------------------------------|-------------------------------|----------|
| [0, 125]                           | 4267.0 (1.79%)                | 960.4    |
| (125, 375]                         | 16737.7 (7.01%)               | 3803.2   |
| (375, 750]                         | 26809.9 (11.22%)              | 6707.0   |
| (750, 1500]                        | 51988.5 (21.76%)              | 14 468.0 |
| (1500, 3125]                       | 88839.4 (37.18%)              | 27 531.1 |
| (3125, 6000]                       | 45929.4 (19.22%)              | 15 679.4 |
| (6000, 9000]                       | 3835.8 (1.61%)                | 2725.7   |
| (9000, ∞)                          | 514.4 (0.22%)                 | 966.9    |
| Total                              | 238922.1 (100%)               |          |

Table 6: Rainy-pixel counts per distance bin for the same distance-to-third-closest-link partition used in Figure 3. For each patch, pixels are classified by the value of  $d_3(p)$ , and the table reports the mean and standard deviation of the rainy-pixel count in each bin across patches.

Using the same definition of the third-closest-link distance  $d_3(p)$  introduced above, and the same distance bins  $B_i$  as in the rainy-pixel analysis, we compute for each patch  $P$ , solver  $s$ , and distance bin  $B_i$

$$m_{s,i}^{\text{nonrain}}(P) \triangleq \text{median}\{\text{AE}_{P,s}(p) : p \in P_{\text{nonrain}}, d_3(p) \in B_i\},$$

and, for fixed  $s$  and  $i$ , the corresponding box-and-whisker plot summarizes the sequence

$$\mu_{s,i}^{\text{nonrain}} \triangleq \{m_{s,i}^{\text{nonrain}}(P)\}_{P \in \mathcal{P}}$$

of patch-level non-rainy-pixel median AE values across patches. Thus the rainy and non-rainy figures use the same distance-bin construction, but they differ both in the pixel set being summarized and in the error metric being aggregated.

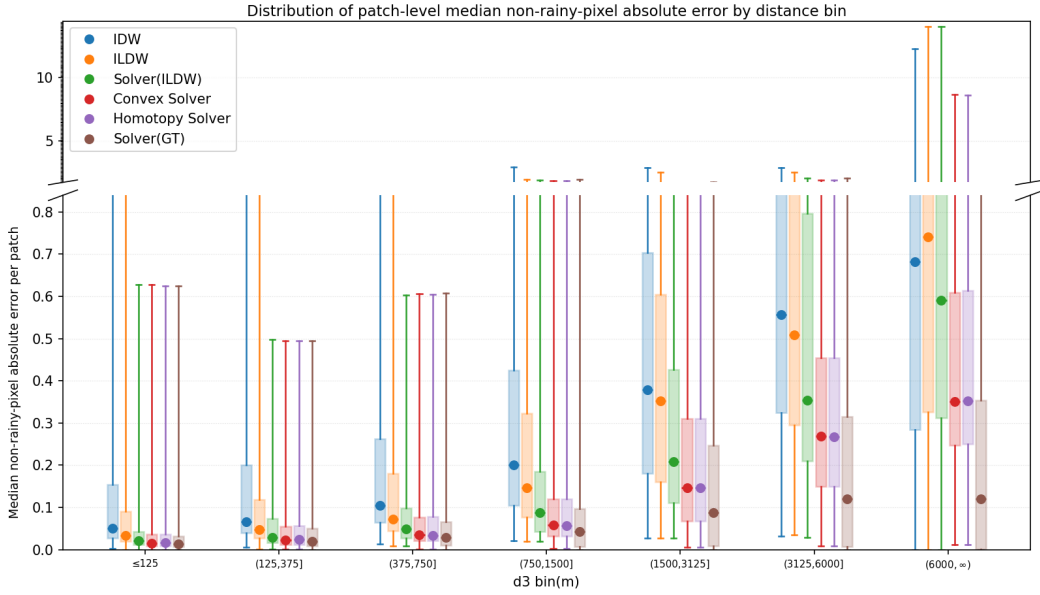


Figure 4: Non-rainy-pixel distance-profile box-and-whisker summary for distance to the third closest link. Each patch contributes one median non-rainy-pixel absolute error value per distance bin, and these patch-level medians are then summarized across patches with a box-and-whisker plot. Non-rainy pixels are those whose ground-truth rainfall is below the configured wet threshold, set here to 0.6 mm/h. The corresponding non-rainy-pixel counts are reported separately in Table 7.

| Distance to third closest link (m) | Non-rainy mean count (% of total) | SD       |
|------------------------------------|-----------------------------------|----------|
| [0, 125]                           | 563.6 (1.84%)                     | 777.6    |
| (125, 375]                         | 2261.1 (7.40%)                    | 3093.5   |
| (375, 750]                         | 3389.1 (11.09%)                   | 4709.5   |
| (750, 1500]                        | 6446.5 (21.09%)                   | 9257.7   |
| (1500, 3125]                       | 10776.2 (35.25%)                  | 15 553.7 |
| (3125, 6000]                       | 6197.2 (20.27%)                   | 8990.0   |
| (6000, 9000]                       | 849.9 (2.78%)                     | 1289.7   |
| (9000, $\infty$ )                  | 85.5 (0.28%)                      | 250.9    |
| Total                              | 30569.2 (100%)                    |          |

Table 7: Non-rainy-pixel counts per distance bin for the same distance-to-third-closest-link partition used in Figure 4. For each patch, pixels are classified by the value of  $d_3(p)$ , and the table reports the mean and standard deviation of the non-rainy-pixel count in each bin across patches.

To make the distance-conditioned summaries easier to interpret geometrically, we show the distance-bin assignment for the largest evaluated patch. This illustrates how the third-closest-link distance partitions a real patch into near-link and low-coverage regions.

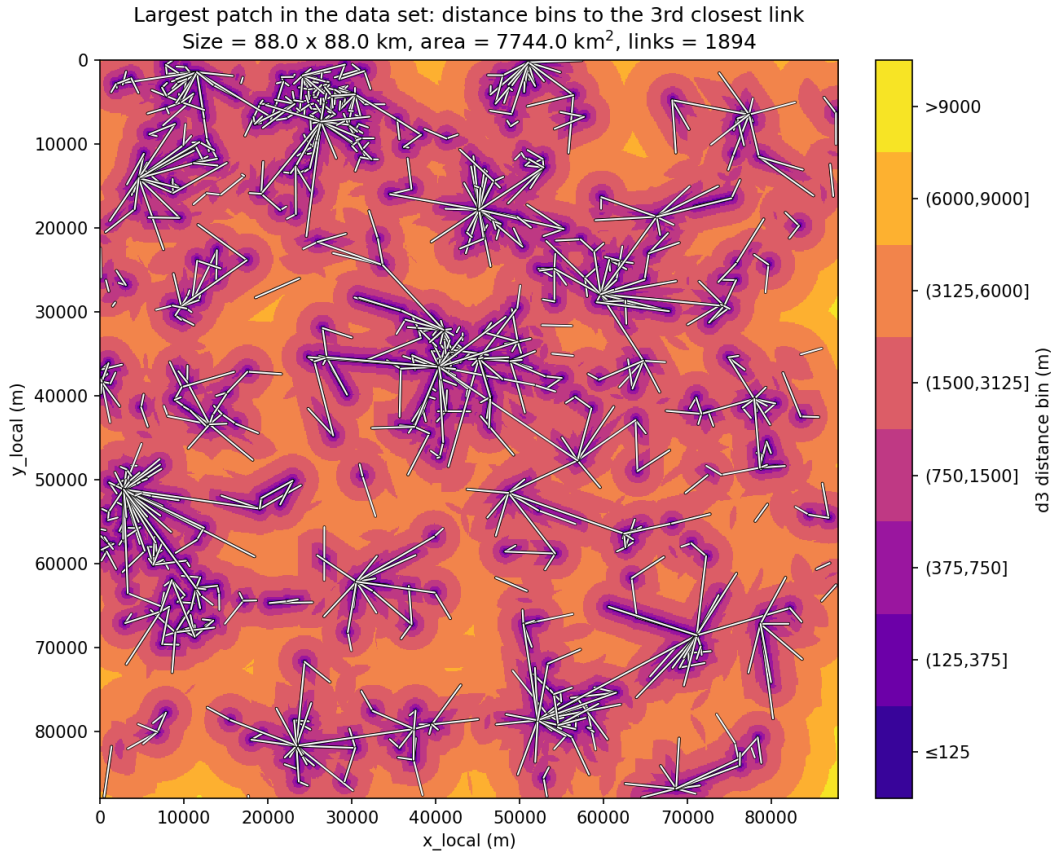


Figure 5: Distance-bin map for the largest evaluated patch in the data set. Each pixel is colored according to the bin induced by its distance to the third closest link, and the complete link geometry is overlaid in white.

### 4.3.3 Wet/Dry Confusion Table

Beyond continuous error magnitudes, we also summarize whether each method preserves the basic wet/dry structure of the rainfall field. Using the same 0.6 mm/h threshold as above, each pixel is classified as wet or dry in both the ground truth and the reconstruction. This gives patch-level true-positive, false-negative, false-positive, and true-negative fractions, summarized in the table below.

| Solver          | TP                | FN                | FP                | TN                |
|-----------------|-------------------|-------------------|-------------------|-------------------|
| IDW             | $0.893 \pm 0.013$ | $0.002 \pm 0.000$ | $0.040 \pm 0.004$ | $0.065 \pm 0.010$ |
| ILDW            | $0.892 \pm 0.013$ | $0.003 \pm 0.000$ | $0.037 \pm 0.004$ | $0.068 \pm 0.010$ |
| Solver(ILDW)    | $0.892 \pm 0.013$ | $0.003 \pm 0.000$ | $0.032 \pm 0.004$ | $0.073 \pm 0.010$ |
| Convex Solver   | $0.891 \pm 0.013$ | $0.004 \pm 0.000$ | $0.023 \pm 0.002$ | $0.082 \pm 0.011$ |
| Homotopy Solver | $0.891 \pm 0.013$ | $0.004 \pm 0.000$ | $0.023 \pm 0.002$ | $0.082 \pm 0.011$ |
| Solver(GT)      | $0.893 \pm 0.013$ | $0.002 \pm 0.000$ | $0.011 \pm 0.001$ | $0.094 \pm 0.012$ |

Table 8: Patch-averaged wet/dry confusion matrix entries at a threshold of 0.6 mm/h, where a pixel is called wet if its rainfall is at least 0.6 mm/h and dry otherwise. For each patch, pixels are first assigned to the four confusion categories defined by the ground-truth and predicted wet/dry labels. The count in each category is then divided by the total number of pixels in that patch, producing a patch-level pixel fraction. The table entries report the mean of those patch-level fractions across the 100 evaluated patches, together with the standard error of the mean (SEM).

False negatives are especially important in this wet/dry view because they correspond to missed rain: pixels that are rainy in the ground truth but classified as dry by the reconstruction. In these rain-event patches, the false-negative fractions are very small for all methods, so the table mainly differentiates the solvers by how much extra wet area they introduce through false positives.

# Conclusion and Outlook

This thesis compared interpolation and optimization-based reconstruction strategies on a 100-patch benchmark built from OPERA rainfall fields and simulated microwave-link attenuations. Across the reported experiments, the optimized solvers achieved much better attenuation consistency than IDW. Convex Solver and Homotopy Solver also improved the map-level RMSE, signed bias, and Pearson correlation summaries relative to both IDW and the ILDW-initialized nonlinear solver. Solver(GT) achieved the strongest overall agreement with the ground-truth rainfall maps, but it is initialized from the ground truth and therefore is not a deployable method. At the same time, the comparison between Solver(GT) and the convex solvers shows that the rainfall field closest to the radar-derived ground truth is not always the field that best matches the link observations under the chosen forward model. ILDW remained a useful link-aware IDW modification: it preserves the simplicity of IDW-style interpolation while making better use of the fact that each CML observation belongs to a full link path rather than only to a midpoint.

Several directions for future work are already visible. One is to move beyond single-time reconstruction and test the current solver on longer event sequences over the same region. Even without changing the objective, this would show whether the method behaves stably over time or produces frame-to-frame artifacts when applied independently to consecutive observations. A second step would be to incorporate temporal structure directly. In meteorology, *optical flow* refers to methods that estimate an apparent motion field between consecutive images (Ayzel et al., 2019). In the present context, such methods could be used to estimate how rainfall structures translate and evolve from one radar scene to the next. That temporal information could then be added to the inverse problem as a penalty that discourages reconstructed rainfall fields from changing in ways inconsistent with the estimated motion field, or it could be used to support short-term rainfall prediction.

One possible extension, related to existing radar-CML data-fusion work, would be to include radar observations directly in the inverse objective, for example through a radar-mismatch term. However, the results above suggest that this should be treated as a formulation question rather than a straightforward add-on: the objective weights would need careful validation because improving the optimized objective is not always the same as improving agreement with the radar-derived ground truth.

Beyond objective design, two broader validation questions remain. First, the benchmark used here samples observed European rain events, but it is not explicitly stratified by rainfall regime; future work should therefore test whether the conclusions hold across labeled rainfall patterns such as convective, stratiform, coastal, or mountainous events. Second, real CML deployment would add time-varying measurement effects, such as changing link availability, wet-antenna attenuation, and baseline drift, which are not present in the simulated-link benchmark and would need to be modeled or filtered before the inverse formulation could be used operationally.

# Appendix

## 6.1 Largest-Patch Link-Network Summary

This appendix summarizes the link-network properties of the largest evaluated patch shown in the distance-bin map. The distance-bin map shows which pixels are close to or far from nearby links, but it does not by itself describe the link network that produced that geometry.

Here, link length refers to the path length of a link, carrier frequency is the microwave-link frequency in GHz, and endpoint degree is the number of links incident to a given endpoint node.

Link length matters because longer links average rainfall over longer paths and can smooth local rainfall variation. Carrier frequency matters because rain-induced attenuation depends on frequency. Endpoint degree gives a compact view of whether the network is mostly chain-like or whether many links meet at shared nodes.

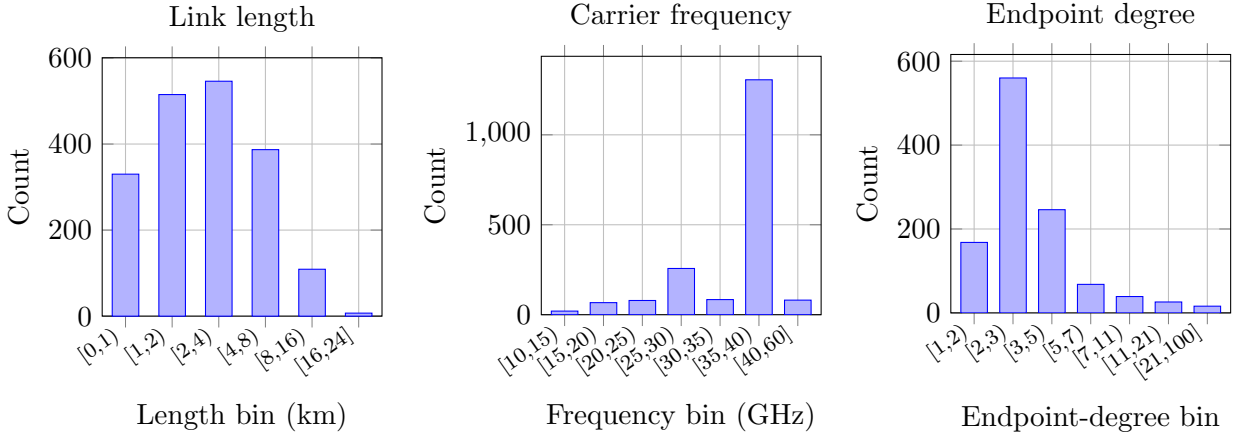


Figure 6: Histogram summaries for the largest patch. The left panel shows link path lengths, the middle panel shows carrier frequencies, and the right panel shows endpoint degrees, where endpoint degree is the number of links incident to a given endpoint node. These distributions provide context for the link geometry underlying the distance-bin map in the main text.

## 6.2 Case Studies

Looking at the case studies that compare GT, IDW, Solver(GT), and Convex Solver, the examples seem to fall into one of two patterns. In the first pattern, Solver(GT) looks very close to GT. In the second pattern, Solver(GT) and Convex Solver look very similar to each other, and both are visibly different from GT.

The first example illustrates the first pattern. For the Black Sea coast, Romania case study, Solver(GT) looks very close to GT, while Convex Solver gives a different reconstruction. Here, Convex Solver reaches a much lower final weighted objective, but Solver(GT) is much closer to the

ground-truth rainfall field by RMSE:

| Solver        | final $J_{\text{weighted sum}}$ | RMSE  |
|---------------|---------------------------------|-------|
| Solver(GT)    | 1.485                           | 0.102 |
| Convex Solver | 0.265                           | 0.702 |

This discrepancy is important: in this case, Convex Solver has a lower final objective value than Solver(GT), even though Solver(GT) is much closer to the ground-truth rainfall field. Thus, minimizing the weighted objective is not the same as minimizing pixelwise reconstruction error to GT. Instead, the objective can prefer a smoother or lower-cost field that fits the attenuation and regularization terms better, even when that field is visually and quantitatively farther from the true rainfall map.

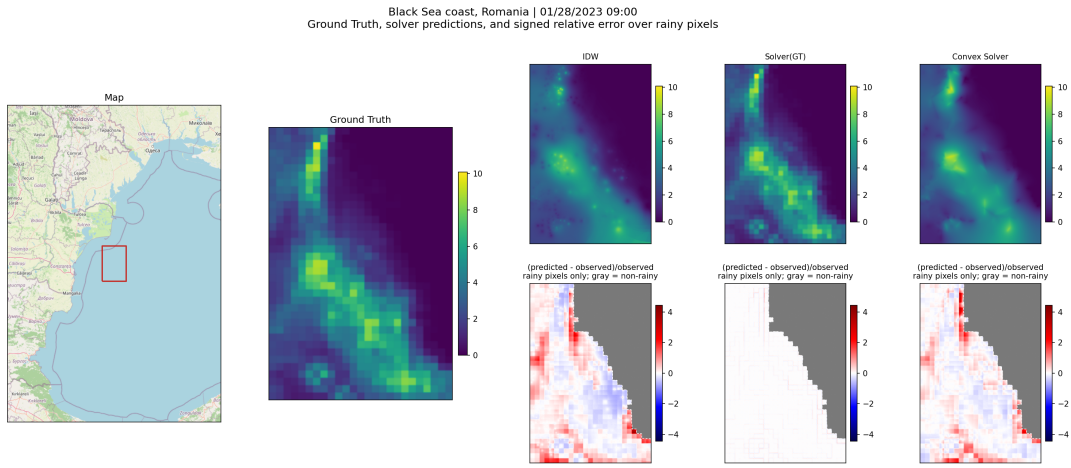


Figure 7: Black Sea coast, Romania case study observed on January 28, 2023 at 09:00. The panel shows GT, IDW, Solver(GT), and Convex Solver.

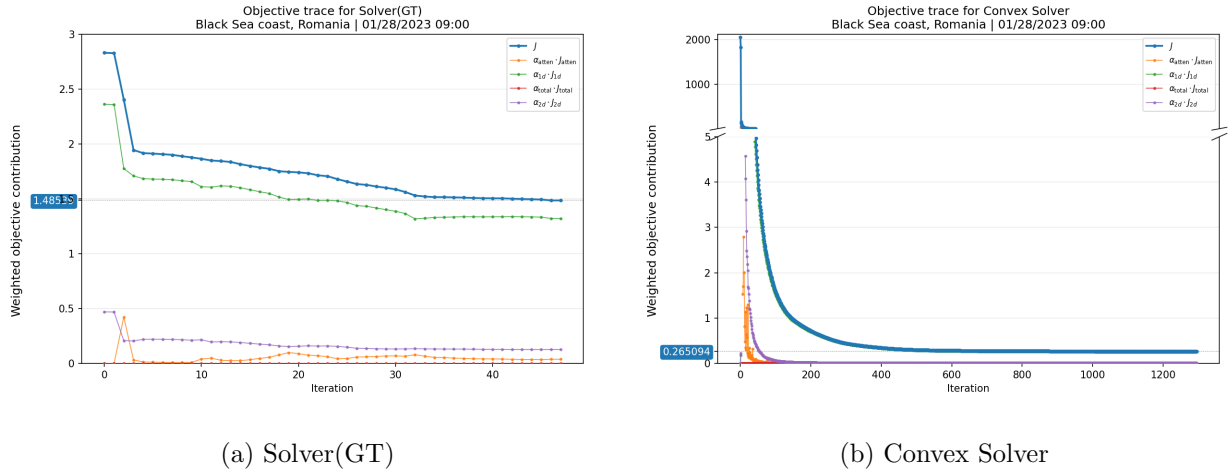


Figure 8: Objective traces for the Black Sea coast, Romania case study.



The traces suggest that most of the objective reduction occurs early. For Convex Solver, the objective is still decreasing after several hundred iterations, but the visible progress after the first few hundred iterations is very small and is driven mainly by the first-discrete-derivative term  $J_{1d}$ . In contrast, Solver(GT) reaches its final behavior after only a few iterations, with one of the stopping tolerances becoming active once the initial objective drop has already occurred.

The second example illustrates the second pattern. For the Northern Norway case study, Solver(GT) and Convex Solver look almost the same in the reconstruction panels:

| Solver        | final $J_{\text{weighted sum}}$ | RMSE  |
|---------------|---------------------------------|-------|
| Solver(GT)    | 0.289                           | 0.552 |
| Convex Solver | 0.290                           | 0.565 |

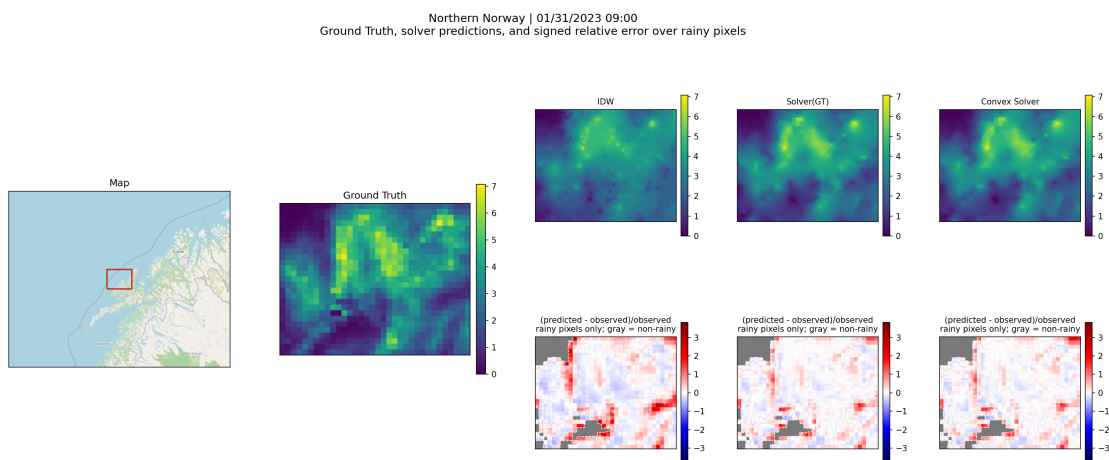
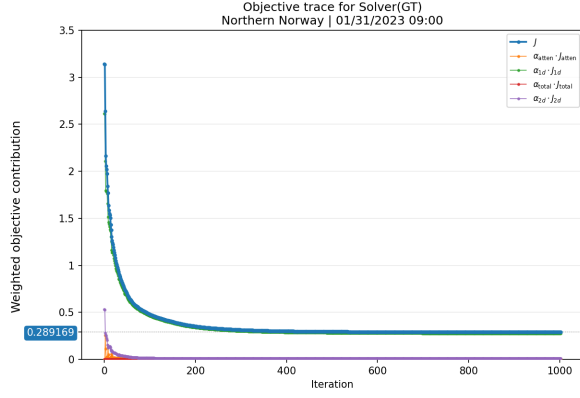
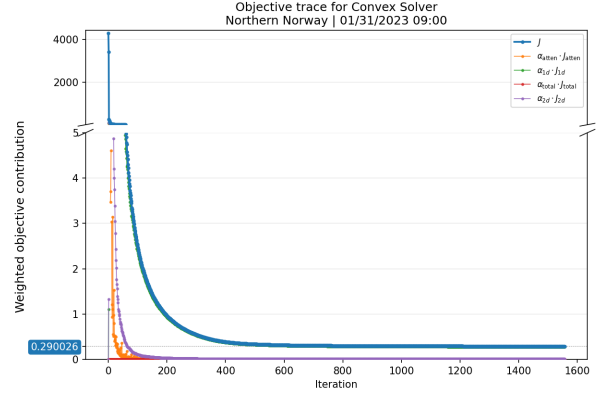


Figure 9: Northern Norway case study observed on January 31, 2023 at 09:00. The panel shows GT, IDW, Solver(GT), and Convex Solver.

Looking closely at the error heatmaps, Solver(GT) and Convex Solver also appear to introduce less extra rainfall than IDW overall, as indicated by the smaller red regions.



(a) Solver(GT)



(b) Convex Solver

Figure 10: Objective traces for the Northern Norway case study.

Similarly to the Black Sea case, the Northern Norway traces show that the first few hundred iterations account for most of the visible objective reduction in Convex Solver, with the remaining gradual decrease driven mainly by the first-discrete-derivative term  $J_{1d}$ .

Taken together, these appendix case studies support the main-text interpretation: Solver(GT) is useful for understanding what the objective can preserve when initialized near the target field, but lower objective values do not always correspond to lower pixelwise error. The examples also show that Convex Solver can either diverge visibly from Solver(GT) or produce nearly the same reconstruction, depending on the patch.

# Statement on the Usage of Generative Digital Assistants

For this thesis, I used OpenAI ChatGPT and OpenAI Codex as generative digital assistants. Their use included support for programming and debugging Python and LaTeX code, rendering and checking report artifacts, converting numerical summaries into LaTeX tables, comparing generated figures and output files, and revising explanatory prose for clarity, structure, and academic style. I also used these tools to help draft intermediate notes, emails, and thesis paragraphs, and to discuss possible interpretations of numerical results and visual case studies.

The tools were used as assistants rather than as independent scientific sources. Mathematical formulations, empirical results, numerical values, figures, citations, and final interpretations were checked against the project code, generated artifacts, local files, and the cited literature. I reviewed and edited the final thesis text and take responsibility for the submitted content.

The research direction, solver design, interpretation of results, and thesis framing were also shaped through supervision by Prof. Guy Even and through project discussions, including meetings or exchanges involving Jonatan Ostrometzky, Christian Sohler, Rex Lei, and Andreas Karenbauer. In broad terms, the work combined supervisory guidance and research discussion with my implementation, experimentation, analysis, and writing, with generative digital assistants used extensively as practical support during coding, analysis, and drafting.

I am aware of the potential risks of using generative digital assistants, including incorrect suggestions, hallucinated claims, and over-polished wording that may obscure uncertainty. For this reason, I used them with caution and critical review.

# References

- Eugene L. Allgower and Kurt Georg. *Introduction to Numerical Continuation Methods*. SIAM, Philadelphia, 2003. doi: 10.1137/1.9780898719154.
- Georgy Ayzel, Maik Heistermann, and Tanja Winterrath. Optical flow models as an open benchmark for radar-based precipitation nowcasting (rainymotion v0.1). *Geoscientific Model Development*, 12(4):1387–1402, 2019. doi: 10.5194/gmd-12-1387-2019.
- Richard H. Byrd, Peihuang Lu, Jorge Nocedal, and Ciyu Zhu. A limited memory algorithm for bound constrained optimization. *SIAM Journal on Scientific Computing*, 16(5):1190–1208, 1995. doi: 10.1137/0916069.
- Christian Chwala and Harald Kunstmann. Commercial microwave link networks for rainfall observation: assessment of the current status and future challenges. *WIREs Water*, 6(2):e1337, 2019. doi: 10.1002/wat2.1337.
- Adam Eshel, Jonatan Ostrometzky, Shani Gat, Pinhas Alpert, and Hagit Messer. Spatial reconstruction of rain fields from wireless telecommunication networks—scenario-dependent analysis of idw-based algorithms. *IEEE Geoscience and Remote Sensing Letters*, 17(5):770–774, 2020. doi: 10.1109/LGRS.2019.2935348.
- Oren Goldshtein, Hagit Messer, and Anat Zinevich. Rain rate estimation using measurements from commercial telecommunications links. *IEEE Transactions on Signal Processing*, 57(4):1616–1625, 2009. doi: 10.1109/TSP.2009.2012554.
- Bin Lian, Zhongcheng Wei, Xiang Sun, and Zhihua Li. A review on rainfall measurement based on commercial microwave links in wireless cellular networks. *Sensors*, 22(12):4395, 2022. doi: 10.3390/s22124395.
- Jack L. McKee and Andrew D. Binns. A review of gauge–radar merging methods for quantitative precipitation estimation in hydrology. *Canadian Water Resources Journal*, 41(1–2):186–203, 2016. doi: 10.1080/07011784.2015.1064786.
- Hagit Messer, Anat Zinevich, and Pinhas Alpert. Environmental monitoring by wireless communication networks. *Science*, 312(5774):713, 2006. doi: 10.1126/science.1120034.
- R. L. Olsen, D. V. Rogers, and D. B. Hodge. The arb relation in the calculation of rain attenuation. *IEEE Transactions on Antennas and Propagation*, 26(2):318–329, 1978. doi: 10.1109/TAP.1978.1141845.
- Jonas Olsson, Laura Horváth-Varga, Remco van de Beek, Maximilian Graf, Aart Overeem, Magdalena Szaton, Vojtěch Bareš, Nejc Bezak, Christian Chwala, Carlo De Michele, Martin Fencl, Jochen Seidel, and Andrijana Todorović. How close are opportunistic rainfall observations to providing societal benefit? *Journal of Hydrometeorology*, 2025. doi: 10.1175/JHM-D-25-0043.1. Early online release.
- Giacomo Roversi, Pier Paolo Alberoni, Anna Fornasiero, and Federico Porcù. Commercial microwave links as a tool for operational rainfall monitoring in northern Italy. *Atmospheric Measurement Techniques*, 13(11):5779–5797, 2020. doi: 10.5194/amt-13-5779-2020.

Ciyou Zhu, Richard H. Byrd, Jorge Nocedal, and José Luis Morales. Algorithm 778: L-bfgs-b: Fortran subroutines for large-scale bound-constrained optimization. *ACM Transactions on Mathematical Software*, 23(4):550–560, 1997. doi: 10.1145/279232.279236.